

MODULE-04

Quantity of Water

- While designing the water supply scheme for the total town or city, it is necessary to determine quantity of water required to various purposes by the city. As a matter of fact, the first duty of the engineer is to determine the water demand of the town and then to find out the suitable water sources from where the demand can be met. For big cities it is not necessary to have only one source of water or one water-works. big cities may draw their water supply from various sources and may have one or more number of water-works distributed in their various localities for the treatment and supply of water.
- While designing the water supply schemes of the town, it is also necessary to determine the total year's demand as well as monthly demand variations in the demand rates.
- Following are the various types of water demand of a city or town:
 - (i) Domestic water demand.
 - (ii) Commercial and Industrial demand.
 - (iii) Demand for public uses.
 - (iv) Compensate losses demand

Domestic water demand

- The quantity of water required for domestic use mainly depends on the habits, social status, climatic conditions and customs of the people. In India on an average, the **domestic consumption of water under normal condition is about 135 litres/day/capita** as per IS: 1172-1171. In **developed countries** this figure may be as high as 350 litres/day capita.

Commercial and Industrial demand

- Commercial building and commercial centres include office building, warehouse, stores, hotels, shopping centres, health centres, schools, temples, cinema houses, Railway and bus stations etc. **The water requirements of commercial and public places may be up to 45 /litres Day/capita.**

Demand for Public uses.

- Quantity of water required for public utility purposes such as for washing and sprinkling of roads cleaning of sewers, watering of public parks, gardens, public fountains etc,

comes under public demand. **To meet the water demand for public use, provision of 5% of the total consumption is made while designing the water works for a city.**

Compensate losses demand

- All the water which goes in distribution pipes does not reach the consumers. Some portion of this is wasted in the pipe lines due to defective pipe-joints, cracked and broken pipes, faulty valves and fittings. Sometimes consumers keep open their taps or public taps even when they are not using the water and allow continuous wastage of water. In some way, some quantity of water is lost due to unauthorised and illegal connections. While estimating the total requirement of water of a town, allowance for these losses and wastages should also be done. **Generally, allowance of 15% of the total quantity of water is made to compensate for losses, thefts and wastage of water.**

Per capita demand

- In community, water is used for various purposes as described above. For the purpose of estimation of total requirements of water, the demand is calculated on an average basis, which is expressed as so many litres/capita/day.
- If Q is the total quantity of water required by a town per year in litres, and the population of the town is P, the per capita demand will be,
- **Per capita demand = $Q/P \times 365$ Litres/day**
- The per capita demand of the town depends on various factors and will be according to the living standard of the public and the number and type of the commercial places in the town etc. For an average Indian town, the requirement of water in various uses is as under

1. Domestic use	135 litres/capita/day
2. Industrial	40 litres/capita/day
3. Public use	25 litres/capita/day
4. Business or Trade	15 litres/capita/day
5. Losses, wastage and thefts	55 litres/capita/day
Total-	270 litres/capita/day

- **Total quantity of water required by the town per day shall be 270 multiplied with the total population of litres/day.**

Drinking Water Standards

The following are the standards of water to be used for domestic purposes:

STANDARDS OF WATER QUALITY

The following are the standards of water to be used for domestic purposes :

Physical

Temperature

- 10°C to 15.6°C

Odour

- 0 to 4 p. value

Colour

- 10 to 20 (platinum cobalt scale)

Turbidity

- 5 to 10 p.p.m. (silica scale)

Taste

- no objectionable taste.

Chemical

Total solids

- upto 500 p.p.m.

Hardness

- 75 p.p.m. to 115 p.p.m. (hardness expressed as Ca CO_3 equivalent)

Chlorides

- upto 250 p.p.m.

Iron and Manganese

- upto 0.3 p.p.m.

pH value

- 6.5 to 8

Lead

- 0.1 p.p.m.

Arsenic

- 0.05 p.p.m.

Sulphate

- upto 250 p.p.m.

Carbonate Alkalinity

- upto 120 p.p.m.

Dissolve oxygen

- 5 to 6 p.p.m.

B.O.D.

- Nil

Biological

B-coli

- No B-coli in 100 ml

Most Probable Number (M.N.P.)

- one number in 100 ml

Radiological

α emitters

- 1 c/litre

β emitters

- 10 c/litre

Water Treatment Process

- The raw water which is found in various natural sources cannot be directly used by the public for the various purposes, before removing the impurities. For palatability the water should be free from unpleasant tastes, odours and must have sparkling appearance. The water must be free from disease-spreading germs. The amount and type of treatment process will depend on the quality of raw water and the standards of quality to be required after treatment.

Objects of Treatment

The main object of the treatment processes is to remove the impurities of raw water and bring the quality of water to the required standards. The objects may be summarized as follows:

- (i) To remove the dissolved gases, murkiness and colour of water.
- (ii) To remove the unpleasant and objectionable tastes and odours 'from the water.
- (iii) To kill all the pathogenic germs, which are harmful to the human health.
- (iv) To make water fit for domestic use as cooking and washing, and various industrial purposes as dyeing, brewing, steam generation etc.
- (v) To eliminate the corrosive properties of water which affects the conduits and pipes.

Layout of Treatment Plant

One complete water treatment plant requires the following process starting from the source of water up to the distribution zone in order of sequence:

- i) Intake work including pumping plant.
- ii) Screening
- iii) Aeration
- iv) Plain sedimentation
- v) Sedimentation aided Coagulation
- vi) Filtration
- vii) Disinfection

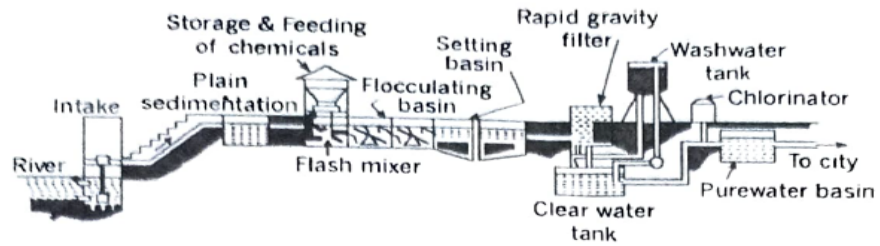


Fig. 10.1. Typical sequence of water treatment plants.

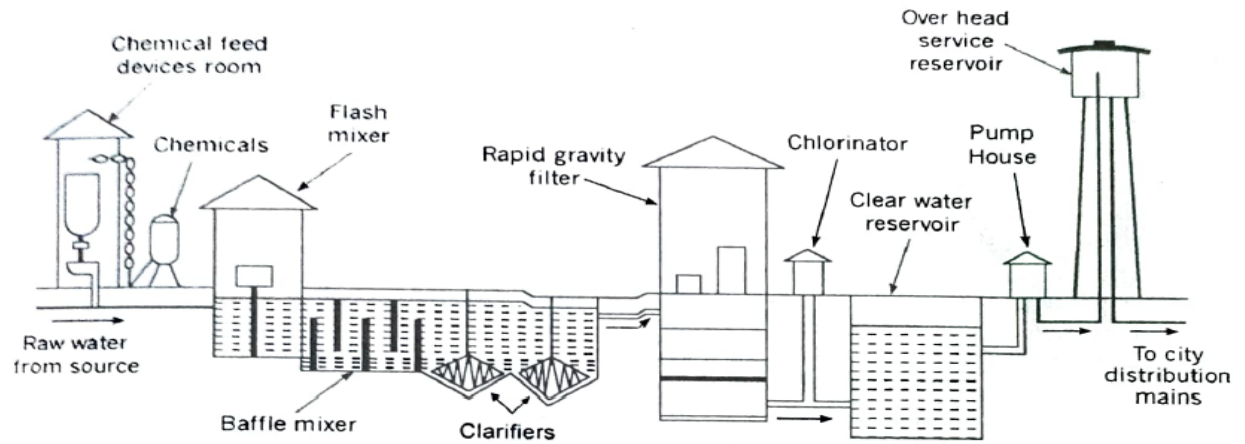


Fig. 10.2. Typical sequence of water treatment units.

Screening

- Fine Screen- Wire mesh of size 10mm
 - Generally avoided as they get frequently clogged requiring frequent cleaning hence increases operational cost.
 - Hence coarse screens are usually used followed by sedimentation and filtration.



Aeration

The process in which water is brought in intimate contact with air to carry out removal of following impurities:-

- It removes dissolved CO_2 , H_2S from water.
- It removes dissolved organic matter from water.
- It removes dissolved minerals like Fe and Mn from water.
- Generally carried out for the treatment of ground water.

Methods of Aeration

- Spray nozzle method-Maximum efficiency
- Removal efficiency:- For CO_2 = 90%
For H_2S = 99%



Cont....

- Cascade aeration-Cascade aerator used.
- Removal efficiency:- For CO_2 = 20-45%
For H_2S = 35%



Sedimentation

- The process of removal of suspended solids from the water is termed as sedimentation. It is based on specific gravity (G) of solids.
- G for inorganic solids= 2.6-2.9 (2.65)
- G for organic solids= 1-2 (1.2)

$G = \frac{\text{Wt. of solids of given vol.}}{\text{Wt. of water of same vol. (} V_s = V_w \text{)}}$
= density of solids/ density of water

Factors effecting sedimentation

- Velocity of flow (v_f) ↓ efficiency ↑
- Viscosity ↓ efficiency ↑
- Size of the particle (d) ↑ efficiency ↑
- Specific gravity (G) ↑ efficiency ↑

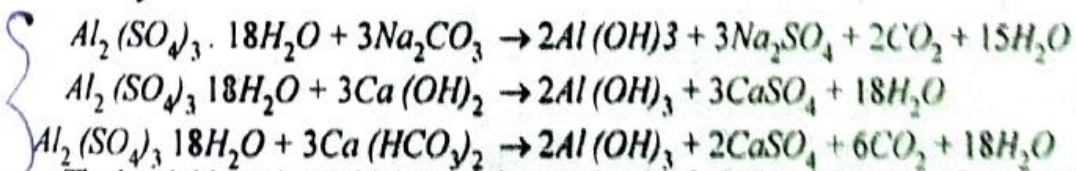
Sedimentation with Coagulation

- Very fine suspended clay particles are not removed by plain sedimentation. Silt particles to 0.06 mm size requires 10 hours to settle in 3 m deep plain sedimentation tank and 0.02 mm particle will require about 4 days for setting. This setting time is impracticable, because water cannot be detained for such a long time. In plain sedimentation tanks detention time of about 2 hours for mechanically cleaned basins and about 6 hours for ordinary tanks. can be allowed.
- In addition to fine suspended matter, water also contains electrically charged colloidal matter which are continuously in motion and never settle down due to gravitational force.
- When water contains such fine clay particles and colloidal impurities, it becomes necessary to apply such process which can easily remove them from the water. After long experience it has been found that such impurities can be removed by sedimentation with coagulation.

12.3 CHEMICAL COAGULANTS

The following are the most commonly used coagulants :

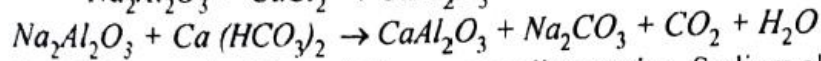
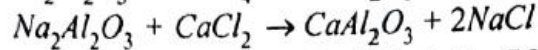
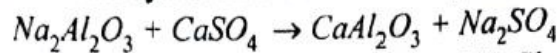
(a) Aluminium sulphate $[Al_2(SO_4)_3 \cdot 18H_2O]$ - It is also called simply as alum. Alum which is available in market, is dirty grey solid in the form of lumps containing about 17% aluminium sulphate. This is the chemical coagulant which is widely used in water treatment plants. Alum reacts in water in the presence of alkalinity, if natural alkalinity is not present sufficient lime is added. The following chemical reactions take place with the various types of alkalinity.



The insoluble and colloidal aluminium hydroxide $[Al(OH)_3]$ forms the floc which removes the fine suspended and colloidal impurities. For best results the pH value of water should be between 6.5 and 8.5. The dose of alum should be 0.03 to 0.13 gm/litre depending on the turbidity of water.

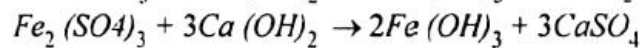
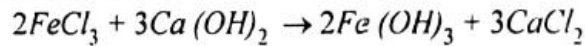
The amount of alum required for coagulation mainly depends on the turbidity and colour of water. The quantity of optimum dose of alum is determined by practical test in the laboratory and it should be adopted in the practice also. Usually the dose of alum varies from 5 mg/litre for relatively clear water to about 25 mg/litre for very turbid waters. The average dose is about 19 mg/litre.

(b) *Sodium Aluminate* [$Na_2Al_2O_3$]. This is an alkaline compound. The best grade it contains Al_2O_3 , 55%; Na_2O , 34%; Na_2CO_3 , 4.5%; $Na(OH)$, 6.3%. This can be used for treatment very easily in the water having no alkalinity. It reacts very quickly and forms the precipitate of aluminium hydroxide. Its chemical equations are as follows :

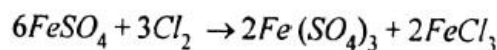


$CaAl_2O_3$ is the required floc, which causes sedimentation. Sodium aluminate does not increase the noncarbondate hardness and it can be easily mixed with lime and soda ash solution. It has the further advantage of removing corrosive qualities of the water. But it is costly due to which it is not widely employed in the water works practice.

(c) *Ferric Cogulants*. Generally ferric chloride ($FeCl_3$), ferric sulphate [$Fe_2(SO_4)_3$] or the mixture of both is used for coagulation purpose. The various chemical reactions which take place are as follows :



(d) *Chlorinated Copperas*. It is a mixture of ferric chloride and ferric sulphate prepared by adding chlorine to a solution of ferrous sulphate in the ratio of 1 part chlorine to 7.9 parts copperas.



It is very good coagulant and requires less amount of alkalinity in the water for floc formation. The produced floc is tough and easily settles due to which only small residue goes in the filters. This coagulant removes colours very well.

Filtration

- Removal of fine suspended particles and flocs from water.
- Helps in removing organic matter, microorganisms & dissolved minerals from water.
- Filtration is done with the help of units termed as filters which are of two types: - Gravity filters and pressure filters.
- Gravity filters are of two types: - Slow sand filters and rapid sand filters.
- Gravity sand filters are those in which head required by the water to flow through the filter medium is provided by height of water itself.

air pressure is maintained on the surface.

ADVANTAGES OF PRESSURE FILTERS

- (i) It is a compact unit and now-a-days automatic pressure filter units have been designed, requiring no manual operation.
- (ii) These units are ideal for small estates and small water works.
- (iii) These filters are very flexible, because the rate of filtration can be changed by change of compressed air pressure over the water.
- (iv) These filters require very small area for their installation.
- (v) Small number of fittings are required in these filters.
- (vi) As the filtered water comes out under pressure no further pumping is required to lift the water.
- (vii) No sedimentation and coagulant tanks are required with these filters.

DISADVANTAGES OF PRESSURE FILTERS

- (i) Overall capacity of the filter is small, though the rate of filtration per *sq.m* is high.
 - (ii) Due to the heavy cost on water treatment, they cannot be used for treating large quantity of water at the water works.
 - (iii) As the process of filtration and back washing is done in the closed tank, proper quality control and inspection is not possible.
 - (iv) The inspection, cleaning, change of filter media, gravel and repair of under drainage system is difficult.
 - (v) Due to circular shape in plan, the design and fixing of back water channels is difficult.
 - (vi) Their efficiency in removal of bacteria and turbidity is poor.
 - (vii) They require additional pumps to pump the water in them.
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Comparison between slow sand and rapid sand gravity filters

	<i>Item</i>	<i>Slow Sand Filter</i>	<i>Rapid Gravity Filter</i>
1.	Area	Requires very large area	Requires small area
2.	Quantity of sand	Requires Considerable quantity of sand	Requires less quantity of sand
3.	Quality of sand	Finer filter media of 0.2 to 0.4 effective size and 2 to 4 uniformity coefficient	Slightly coarser filter media of 0.36 to 0.6 effective size and 1.2 to 1.8 uniformity coefficient

FILTRATION

Part-I / 185

	<i>Item</i>	<i>Slow Sand Filter</i>	<i>Rapid Gravity Filter</i>
4.	Quality of raw water	It may not be treated with chemicals, but should not have turbidity more than 50 p.p.m.	Treatment with chemicals is essential.
5.	Flexibility in operation	Not possible	Possible
6.	Rate of filtration	100-180 litres/m ² /hour	4,000-5,000 litres m ² /hour
7.	Size of one unit	30 m × 60 m	6 m × 8 to 8 m × 10 m
8.	Distribution	Uniform	Smaller at top and coarser in bottom
9.	Underdrainage system	Open jointed pipes or drains covered with blocks	Manifold and pipe laterals. Vitrified tile Blocks ; the Wheeler Filter Bottom ; the Porous Plate Bottom Concrete Ridge and Valley Bottom etc.
10.	Period of Cleaning	1 to 3 months	24 to 48 hours

11.	Method of cleaning	Scrapping 2-3 <i>cm</i> sand from the surface and replacing it with new sand.	By back washing with water under pressure, with or without compressed air agitation before washing with water
12.	Skilled Supervision	Not required	Most essential
13.	Loss of head	15 <i>cm</i> to 75 <i>cm</i>	2 <i>m</i> to 4 <i>m</i>
14.	Penetration of suspended impurities	Very small, only a dirty layer is formed at the surface	2 <i>m</i> to 4 <i>m</i> Very deep
15.	Amount of water required for washing	0.2 to 0.6% of filtered water	2 to 4% of filtered water
16.	Overall cost of unit	More, because large land and much quantity of materials are required	Cheap and economical
17.	Cost of Maintenance	Small	More
18.	Efficiency	Efficient in removal of bacteria and suspended matter	Cannot remove all bacteria, disinfection necessary. Removes colour, odour and taste.

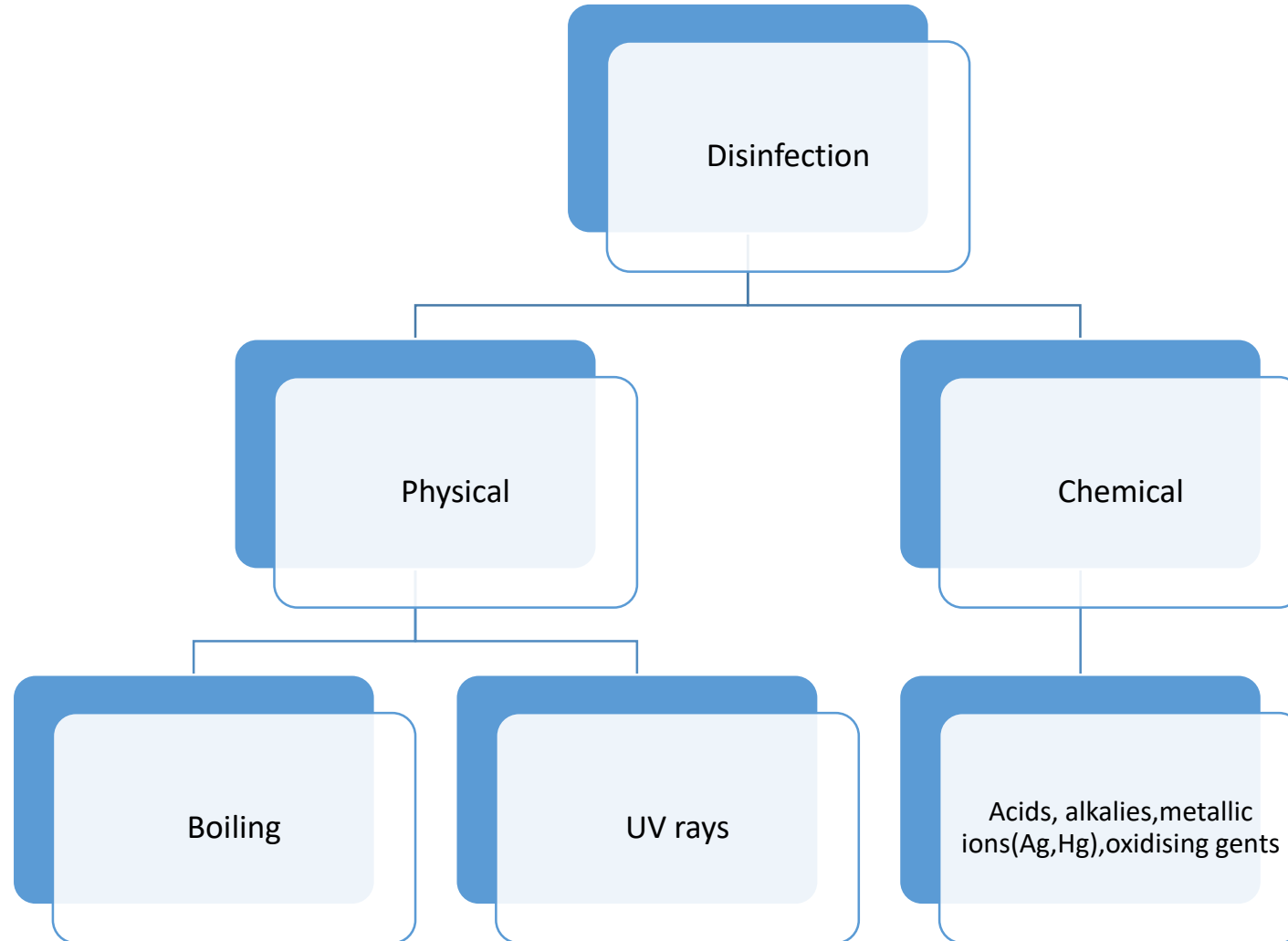
Disinfection

- Process of removal of disease causing microorganisms from water is called disinfection.

Requirements of good disinfectant

- They should destroy all the diseases producing microorganism and make it safe for use.
- They should not take more time in killing pathogens, but do their task within the required time.
- They should be economical and easily available.
- After treatment the water should not become toxic and objectionable to the user.
- Their dose should be such that it leaves some residual for protection against future contamination.

Methods of Disinfection



Minor methods

- **Boiling:** The water can be disinfected by boiling for 1 to 20 mins. This process only can kill the existing germs but do not provide protection against any future protection.
- **UV rays:** These are invisible rays having wave length of 1000 -4000m. UV rays are highly disinfectant and kill the pathogens. After removing the turbidity and colour disinfection with the UV rays are done. As no chemicals are used it do not impart any taste or odour in water.

Treatment with Ozone-When added to water it disassociates into molecular & nascent oxygen which is strong oxidizing agent that carries removal of both organic matters and microorganisms from water but is highly unstable & nothing remains in water till it reaches the distribution systems thereby it doesn't safeguard water against future contamination.

Normal dose= 2-3ppm

Costlier than chlorination but imparts pleasant taste to water.

It destroys the cell wall of the micro organisms to carry out disinfection.

Treatment with Potassium Permanganate(KMnO₄)

Popular disinfectant used for treatment of well water supplies in villages which are contaminated to lesser extent.

It removes both organic matter and micro organisms from water.

When added in water it imparts pink colour which disappears if OM and microorganisms are present. On adding it more if the pink colour persists which indicates the complete removal of OM & microorganisms.

Water is not used for next 48hrs for pink colour to subsidize.

Normal dose=1-2ppm at contact period=4-6hrs.

% removal=98% of bacteria and 100% cholera bacteria.

Major methods of treatment

- Treatment with chlorine: Chlorine is universally used for the disinfection of the public water supply. It is cheap, reliable, easy to handle, easy to measure the residual amount. If used in greater amount it imparts bitter taste.
- When chlorine is added to water, it forms hypochlorous acid or hypochlorite ion.
- $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{OCl}^-$ free available chlorine

Types of chlorination

1. **Plain Chlorination:** This term is used to mean that only chlorination is used no other treatment is used for treatment of water. This treatment can be only in the case of clear water (with turbidities less than 20-30 mg/l) obtained from lake, reservoirs, well. It can also be used during emergency like during war water can be supplied in army troops after this treatment.
2. **Pre chlorination:** In this process chlorine is added in the water before filtration or even before coagulation. Prechlorination is generally done when the quantity of the water is poor with respect to the organic matter and microorganism present in it.
Its normal dose is 5-10 mg/l, if added during coagulation such that 0.1-0.5mg/l reaches upto filtration.

3. Post Chlorination: If chlorination is the last treatment given to the water, it is termed as post chlorination.

Its dose is such that 0.2mg/l of residual is ensured at a contact period of 10 minutes.

4. Double Chlorination: When both the pre and post chlorination is done, it is referred to as double chlorination. Post chlorination is always used, but pre chlorination is used when the water is highly turbid and contaminated.

5. Super Chlorination: When excess chlorine in the range of 5-15mg/l is added in the water at the time of epidemic so as to ensure residual of 1-2mg/l, it is referred as super chlorination.

6. De Chlorination: The process of removal of excess chlorine from the water is termed as dechlorination.

It can be achieved by the addition of any of the following reagents – SO_2 gas, Activated carbon, Sodium Sulphite (Na_2SO_3), Sodium bisulphite (NaHSO_3).